Teaching with this workbook

This 50-page companion follows the current five-module course and all 15 named sections. It provides 30 selected source-exact MCQs, paper applications and the 12 current folio purposes. It does not replace the full online question bank, the original project drawing, current school safety procedures or teacher demonstrations.

The student PDF and native editable Word edition carry the same teaching and response spaces. Print A4 double-sided on the long edge. The original landscape drawing is rotated on page 8 so students can turn the book to inspect it. Written dimensions retain authority after resizing.

Use classroom discussion to establish the five success-criteria categories on page 12. Practical dimensions, stock, joints, assembly, laser workflow, machine settings, electronics, pins, voltage, strip type, libraries, upload procedures, final code, tests and submission details must be supplied through authoritative teacher instruction. Nothing in this book grants equipment permission.

The paper algorithms on pages 33–35 are fictional language-neutral logic exercises. Their states and colours are teaching examples, not project settings. Generated images are illustrative; the full source plan controls practical form.

Assessment is formative unless the teacher issues separate formal instructions. The notes below give expected reasoning and valid alternatives rather than a fabricated marking scheme.

Source scope and unresolved details

Source snapshot: programmable-light-guided-course commit recorded in the build source manifest, retrieved 10 September 2026. Current guided/data.js and folio.html control the five-module sequence and twelve folio records. Older source-note route maps are historical. Thirty questions are selected from the current 150-question bank, retaining stems, option order and correct feedback.

The original one-page Programmable Lamp drawing is titled by Josh McKenzie, dated 24/02/2025. Its master PDF is preserved unchanged. It does not explicitly state units, material allocations or tolerances. Feature interpretation and the safe methods for small components need teacher checking; never infer safe disc-sander use from an isolated part shown in a picture.

The two-layer practice resource allows a maximum 80 mm×50 mm key tag. That limit is not a lamp dimension. The original image-search worksheet’s eleven prompts are adapted into the transferable observation/source-record activity on page 14 and the six-idea folio; a teacher may provide additional printed references.

Timber-production.pdf page 3 asks for a thirteen-stage order from an unidentified video. Page 27 is therefore a clearly labelled teacher-led extension: no fabricated sequence or answer key is supplied. Provide the matching source before using it. Core material and sustainability learning is fully taught on pages 23–26.

What-is-coding.pdf page 1 supports the vocabulary. Its unrelated second-page AI-video questions are excluded consistently with the current named course scope and upstream visual audit. Presentations and media remain teacher-led supplements. Exact practical and formal requirements follow the course teacher-confirmation register.

Guidance for student pages 3 to 5

Student page 5 • Accept specific hazards linked to plausible harm and suitable controls from the current school procedure. For example, a loose object in a route can cause a trip; removing it to the designated storage area controls that hazard. Do not award completeness for repeating "be careful".

Student page 5 • A justified priority refers to the possible harm and exposure, and identifies the safe response. The teacher checks the local context; students do not enter an unsafe area to gather evidence.

Guidance for student pages 6 to 8

Student page 6 • Five further distinct, source-supported workshop observations complete the ten-hazard investigation across pages 5–6. Check controls against the school procedure, including authorised boundaries and reporting.

Student page 6 • Stop the activity safely within the student’s training, warn others as appropriate, and tell the teacher. Do not attempt unauthorised repairs or continue until the teacher resolves the issue.

Student page 6 • Teacher supplies current local details. This record documents discussion; it is not itself a safety test pass, machine permission or formal sign-off.

Guidance for student pages 9 to 11

Student page 9 • Programmable Lamp; Josh McKenzie; 24/02/2025; overall 1:2; two details labelled 1:1. Preserve the displayed information; do not infer units from typical workshop practice.

Student page 9 • The visible labels are 180 and 75 in the upper-left view; 59 and 25 in the upper-middle profile; 59 and 19 in the lower-left pictorial; and 59, 10, 30 and 5 in the cutting detail. Units are not explicitly stated. Teacher verifies each feature relationship from the full plan before practical use.

Student page 9 • Examples: written unit, material allocation, tolerance, stock or an unclear feature relationship. The question must be specific enough for teacher clarification.

Student page 10 • 1.1: B. The project brings together physical construction, lighting, electronics, code and design.

Student page 10 • 1.11: B. A hazard is something with the potential to cause injury or other harm.

Student page 10 • 1.21: B. The measurement worksheet identifies millimetres as the precision unit used by the laser cutter.

Student page 10 • 8 cm =80 mm;0.25 m =250 mm. Multiply centimetres by 10 and metres by 1000. Values here are conversion exercises, not project dimensions.

Student page 11 • 1.9: D. A useful criterion states something observable that can be checked when the solution is evaluated.

Student page 11 • 1.19: D. An SOP supports safe work but must be used with training, permission and local controls.

Student page 11 • 1.28: D. This matches the numbered sequence written on the authorised drawing.

Student page 11 • A preview may have been resized. Use the written dimension, associate it with the correct feature and view, and resolve missing units or practical details with the teacher before work.

Guidance for student pages 12 to 14

Student page 12 • Look for a purpose connecting physical design, light, electronics and intended behaviour. User/setting are student proposals unless the teacher provides them.

Student page 12 • Each confirmed criterion should be observable and linked to a suitable evidence method. Categories are source supplied; exact criteria require class/teacher agreement. Avoid manufactured performance thresholds.

Student page 12 • A personal goal targets the student's learning, such as accurately interpreting a drawing or explaining a program trace, with evidence of progress. It is distinct from a product criterion.

Student page 14 • Possible features include the repeated angular lines, contrast between transparent panel and opaque base, or restrained colour contrast. A developed response changes or combines a feature for a stated purpose. Do not infer unseen construction or operational performance.

Student page 14 • For the supplied image, record original generated concept, created for this workbook using image_gen; actual pixel dimensions are available in the image record. For an independent research image, preserve its real creator/site/URL and permissions. Unknowns remain identified, not guessed.

Guidance for student pages 15 to 17

Student page 16 • Accept a coherent base outline with dimensions no larger than 80×50 mm, clearly identified as the key-tag practice. Practical hole/feature suitability is teacher checked. The workbook box is not a full-size manufacturing template.

Student page 16 • The detail layer should be distinguishable from the base and remain readable. An explanation should identify deliberate open/closed paths. Do not prescribe cut/engrave colours or laser settings.

Student page 16 • Filename and export format reflect the actual teacher instructions. Reflection identifies a specific improvement to path clarity, layer organisation or simplicity.

Guidance for student pages 18 to 20

Student page 18 • Idea 1: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 18 • Idea 2: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 18 • Idea 3: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 19 • Idea 4: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 19 • Idea 5: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 19 • Idea 6: inspect a recognisable proposal with annotations explaining the explored feature and difference. Across pages 18–19 require six distinct, numbered, explained alternatives; no requirement to manufacture all six.

Student page 20 • Compare at least three of the six ideas using the same relevant confirmed criteria. Credit a reasoned trade-off, not an unexplained popularity vote or decorative score.

Student page 20 • Refinement must respond to evidence, a criterion or feedback. Assess annotation clarity and reasoning. Proposed changes to the practical form require teacher agreement against the original plan.

Student page 20 • A justified choice refers to the comparison and names one unresolved technical or practical question. Distinguish a preference from a requirement; do not claim a concept is approved for manufacture.

Guidance for student pages 21 to 23

Student page 21 • 2.1: B. Research provides visual evidence that can support idea generation, comparison and explanation.

Student page 21 • 2.11: A. Vector designs are built from editable paths rather than relying on a fixed pixel grid.

Student page 21 • 2.21: A. Iteration uses repeated cycles of development to move towards a more suitable design result.

Student page 21 • Pixel dimensions indicate available image detail; they do not create editable paths, organise layers, establish manufacturability or grant reuse permission. A vector design must be deliberately developed and checked.

Student page 22 • 2.9: C. Practice with outlines, layers and simple shapes builds transferable vector-design skills.

Student page 22 • 2.19: B. A detailed pixel image may be useful research, but a vector design still needs clear, intentional lines and curves.

Student page 22 • 2.28: C. The two statements serve different purposes within the project.

Student page 22 • Example criterion proposal: the chosen symbol can be identified at the agreed viewing conditions; teacher confirms it. Example personal goal: improve annotation clarity, evidenced by labelled sketches. A product criterion judges the solution; a goal judges individual learning.

Guidance for student pages 24 to 26

Student page 25 • Timber suitability needs confirmed properties, stock and purpose; sourcing needs origin, management/regeneration and reliable traceable information; acrylic suitability needs the approved specification/design and teacher guidance. Appearance alone does not establish these claims.

Student page 25 • Radiata pine and 3 mm clear acrylic are named project materials. A defensible response connects a relevant confirmed property or requirement to purpose and notes an unknown such as stock, origin or final detail suitability. Do not assert all softwoods are cheap, easy or sustainable.

Guidance for student pages 27 to 29

Student page 27 • Media-dependent extension: verify all thirteen positions against the specific teacher-supplied video, not a generic sawmill sequence. The worksheet labels include merchandising/infeed/sorting/debarking/sawing/drying/planing/trimming/stacking/branding, but their exact order cannot be authenticated from the available snapshot. Do not manufacture an answer key.

Student page 27 • Record the actual video title/source and any missing or unclear stages. An unverified ordering remains incomplete as an observation task; it is not represented as a confirmed production method.

Student page 28 • 3.1: B. These terms classify timber by botanical origin rather than guaranteeing its physical hardness.

Student page 28 • 3.11: B. Native forests are defined by native tree species occurring within their natural location and range.

Student page 28 • 3.21: B. A production plan should begin with verified drawing information rather than habits or assumptions from unrelated work.

Student page 28 • Softwood is a botanical grouping. Cost, workability and suitability depend on the actual timber, condition, stock and intended use, supported by evidence. Radiata pine is named by the project source; the label alone is not a complete selection rationale.

Student page 29 • 3.9: C. The sources do not prove that every softwood shares the same cost or workability.

Student page 29 • 3.19: C. Forest type alone does not determine sustainability; management evidence and impacts must also be considered.

Student page 29 • 3.28: D. A dated, stage-linked photograph can show what was completed and when the checkpoint occurred.

Student page 29 • Order: Cut First and Sand; Cut Second and Sand; Cut Third. They do not supply complete procedures, teacher permission, machine settings, units, stock suitability or every feature relationship. The teacher controls safe methods and checks.

Guidance for student pages 33 to 35

Student page 33 • The whole blue/amber pair occurs three times: blue, amber, blue, amber, blue, amber. “Twice more” means two additional full repetitions after the first pair, not two records in total.

Student page 33 • The ordered set-and-record pair forms a sequence. The named colour acts as changing information, represented by a variable. Repeating the complete sequence is a loop; three pairs produce six recorded outputs.

Student page 34 • A: yes, amber; B: no, blue; A: yes, amber; C: no, blue. The rule tests equality with A, so every other supplied state follows the no branch.

Student page 34 • The condition is whether the state is A, not whether it is B. C fails that condition and therefore uses the no branch, which gives blue. Tracing the stated rule avoids adding an unstated third path.

Student page 34 • A state was supplied on paper to teach conditional reasoning. It does not identify a real input device, circuit, programming language or project control rule.

Student page 35 • Accept a clear proposed behaviour with a coherent order. Values or timings must be labelled proposals, not approved settings. A student need not invent a sensor or input.

Student page 35 • Look for ordered steps, clear branches and repetition where needed. Arduino Uno is the named controller; RGB LED strip is the named light output. Any unspecified input/control remains “Teacher to confirm”. This must not become an invented circuit.

Student page 35 • A test case states a supplied condition/state and a predictable outcome using the student’s algorithm. The unresolved detail may concern the actual input, strip type, wiring, electrical requirements, language or approved control rule.

Guidance for student pages 36 to 38

Student page 36 • 4.1: B. Coding communicates instructions that a digital system can follow to produce an intended behaviour.

Student page 36 • 4.11: A. A variable allows changing information to be identified and used within an organised digital process.

Student page 36 • 4.21: B. A system combines related parts so they contribute to a shared purpose or intended result.

Student page 36 • An algorithm is an ordered description of behaviour; an Arduino Uno is a physical controller. The algorithm may be expressed in ordinary language before it is translated into the confirmed programming language. Instructions are not a physical wire or component.

Student page 37 • 4.9: C. Sequencing arranges instructions in the order needed to produce the intended result.

Student page 37 • 4.19: D. A language-neutral plan should identify the changing information, repetition and decisions that shape the intended behaviour.

Student page 37 • 4.28: D. A generic input or control label communicates the required role without inventing a device.

Student page 37 • It can show a conceptual relationship between controller and light output. It cannot establish a direct electrical connection, signal type, power arrangement, pins, driver requirement or safe wiring. Those need the approved technical configuration.

Guidance for student pages 39 to 41

Student page 41 • The criterion and method must be confirmed before practical testing. No numerical threshold, electrical test or load is prescribed by this workbook. A paper trace may be used where implementation is not yet authorised.

Student page 41 • Evidence is an actual observation, not the hoped-for outcome. A cause is a hypothesis until supported. Record the approved change and retest, including a result that does not improve performance.

Student page 41 • The conclusion must be limited to the aspect and conditions checked. Improvement needs evidence of a closer match to the criterion; changing several things at once weakens explanation of cause.

Guidance for student pages 42 to 44

Student page 42 • Accept dated, specific records linked to the supplied plan or named teacher instruction. The observation describes actual evidence, including problems and corrections. Do not reward an invented perfect sequence or fabricated photographs.

Student page 42 • A useful account identifies the problem, the approved response and the observed result or retest. If the problem remains unresolved, state that honestly; no unsupported diagnosis or repair instructions are required.

Student page 43 • 5.1: B. Integration considers how design, materials, electronics and code contribute to the same intended solution.

Student page 43 • 5.11: B. Testing requires a known expectation, an observed result and a clear comparison between them.

Student page 43 • 5.21: D. Evaluation connects a judgement with relevant evidence and a clear basis for comparison.

Student page 43 • Several changed factors confound the comparison. An approved controlled check of one aspect, with prediction and before/after evidence, gives a clearer basis for locating a cause. A different result is not automatically an improvement.

Student page 44 • 5.9: D. Traceable communication shows the evidence and reasoning behind each design decision.

Student page 44 • 5.19: B. Improvement must be supported by comparison evidence rather than preference or the fact that a change occurred.

Student page 44 • 5.28: C. These forms of evaluation have different subjects: student development and product performance.

Student page 44 • Example: “I can now distinguish a variable from a loop; my corrected trace on page 33 shows changing information and three repeated pairs.” The evidence must be actual student work when submitted. A product claim instead compares product evidence with its criterion.

Guidance for student pages 45 to 47

Student page 45 • Assess each of the five agreed categories: design, function, construction/build quality, electronics/programming and aesthetics. The student should cite earlier evidence and explain why it supports the judgement. Missing criteria or unobserved tests remain limitations, not invented passes.

Student page 45 • A strong response links a particular feature or performance result to the confirmed criterion, names the observation or record, and explains the relationship. A photograph can support visible appearance but not every functional claim.

Student page 46 • Accept a specific limitation and proportionate proposal supported by reasoning. A predicted effect is not a proven result. Technical changes remain subject to the approved plan and teacher review.

Student page 46 • Connect a sustainability judgement to evidence about origin, management, regeneration or observed resource use. The current snapshot does not establish the supplied timber's certification or complete lifecycle effects.

Student page 46 • An honest comparison cites the goal and specific student evidence, acknowledges a remaining need and states a practical next learning action. This is individual reflection, not a formal grade.

Student page 47 • Tenon saw: controlled hand sawing; tri-square: marking/checking right angles; pencil: visible marks; steel rule: measuring; soldering iron: heating approved solder joints; helping hands: holding a small work assembly; solder: joining medium in an approved connection; laser cutter: teacher-controlled processing of an approved compatible design/material. Do not prescribe settings, circuitry or a practical sequence.

Student page 47 • Check the actual classroom item or an accurately labelled source. A tri-square has a fixed square stock/blade relationship; do not relabel a combination square as a tri-square. The chosen identifying feature must be visible and relevant.

Guidance for student pages 48 to 50

Student page 48 • Radiata pine: named timber; screws: named fasteners; clear acrylic: named transparent sheet material of stated 3 mm thickness; Arduino Uno: controller; RGB LED strip: light output; jumper wires and USB cable: named connection materials. Exact allocations, stock, fitting, wiring, electrical requirements and connector ends are not fully specified. Preserve the source cable wording without treating it as a complete connector specification.

Student page 48 • For example, acrylic contributes to physical/visual design while the RGB strip provides light output. A useful response distinguishes roles and explains their relationship without inventing an actual connection or arrangement.

Student page 50 • Readiness is evidence based, not a formal mark. Record incomplete or teacher-dependent items honestly, especially the optional matching-video task on page 27. The source booklet’s electrical, practical and submission details remain teacher controlled.

Student page 50 • Teacher supplies the actual submission instructions. The paper book or device-local folio does not automatically submit work. Keep source/photo/code files required by the teacher and verify the real submission method.